
WAYPOINT STUDIO

Full Platform Engineering Audit Report

Stability, Hardening & Architecture Review

REPORT DATE

12 July 2026

DOCUMENT

PLATFORM-AUDIT-2026-07

BRANCH

main

AUDIENCE

Senior engineering handoff

This PDF is a presentation of the Markdown source. Report content is unchanged.

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Commits created during the broader July platform arc

Files changed in hardening commit 0b74836 (19 files)

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Risks

Recommended Next Sprint

Single highest-value next sprint: National-mode honesty + weather request coalescing + dashboard favorites cleanup

Why this over alternatives

Success criteria (suggested)

Overall Assessment

Appendix A — Verification commands

Appendix B — Related documents

Waypoint Studio — Full Platform Engineering Audit Report

Document: docs/PLATFORM-AUDIT-2026-07.md
Sprint: Full Platform Audit, Stability, and Hardening
Audit date: 2026-07-12
Repository: bfree7885/waypoint-studio (local path waypoint-scenes)
Branch: main
Hardening commit: 0b748364f4ece7c5ffef4ecc54fdf4e078ca9bbe (0b74836)
Preceding product commits in scope:

Commit	Summary
3a28bc1	Unified Platform Foundation
3900125	Shared Knowledge Platform
f26eb40	Fieldry Life List MVP
0b74836	Platform stability hardening (this audit's primary fix commit)

Authors of this report: Platform audit / hardening sprint (lead architect + reliability + QA synthesis)
Audience: Senior engineers inheriting Waypoint Studio maintenance and roadmap decisions

Executive Summary

Overall assessment of the repository

Waypoint Studio is a multi-application, shared-kernel frontend platform for outdoor intelligence, observation, photography coaching, and domain foundations (sheds, tea, radio terrain, wine). The architecture is coherent: a design-system loader (wds.js / wds-platform.js) provides Outdoor Intelligence Platform (OIP), location, weather, Waypoint Observation Schema (WOS), Knowledge Platform, and shared shell/stores. Product apps consume those APIs rather than inventing parallel stacks—with important historical exceptions (Photo Coach dual storage, kiosk dual boot lineage, dashboard favorites dual-read).

The July 2026 audit found that the platform was **feature-rich and largely shippable**, but several **initialization and location-correctness defects** could produce silent wrong geography, broken Fieldry/ForageCast boot, or wiped WOS extension APIs depending on script order. Those defects were fixed in 0b74836 .

Major improvements made

1. **Lazy app boot** for Fieldry and ForageCast (wait for `WDS.appBoot` instead of capturing `null` once).
2. **WOS module merge** so `normalizeObservation` APIs and `observations.extensions` coexist across load order.
3. **Knowledge module merge** so `search` / `related` survive core reload.
4. **Location hardening**: reject only the engine publish point (US center), not an entire Kansas bounding box; stop silent Pike County substitution on OIP failure; require explicit `allowDefaultLocation` for defaults.
5. **XSS hardening** on species profile URL ids and Scenes photography attribute ids.
6. **CI expansion** to run foundation, Knowledge, Fieldry, hardening, photographer profile, and coaching unit tests; expanded headless smoke route list.
7. **Documentation**: this audit, storage inventory, unified-platform cross-links.

Current production readiness

Surface	Readiness
Outdoor Intelligence Dashboard	Production-capable; location validators green; engine data guarded from user weather/river paths
Kiosk	Production-capable for display; still consumes engine artifacts under guards
ForageCast	Live; boot race fixed
Fieldry	Usable Life List MVP; boot race fixed; WOS + Knowledge wired
Waypoint Scenes / Photo Coach	Live public products; dual storage debt remains
Sheds / Steepleaf / SignalTerrain / Savant	Architecture-ready foundations, not finished products
Terrainbound	Correctly retired → Fieldry redirect
Future data / marketplace	Disabled by design

Verdict: Safe to keep shipping on `main` for the public portfolio. Not “finished platform”—foundations and Photo Coach storage consolidation remain intentional follow-on work.

Overall platform health

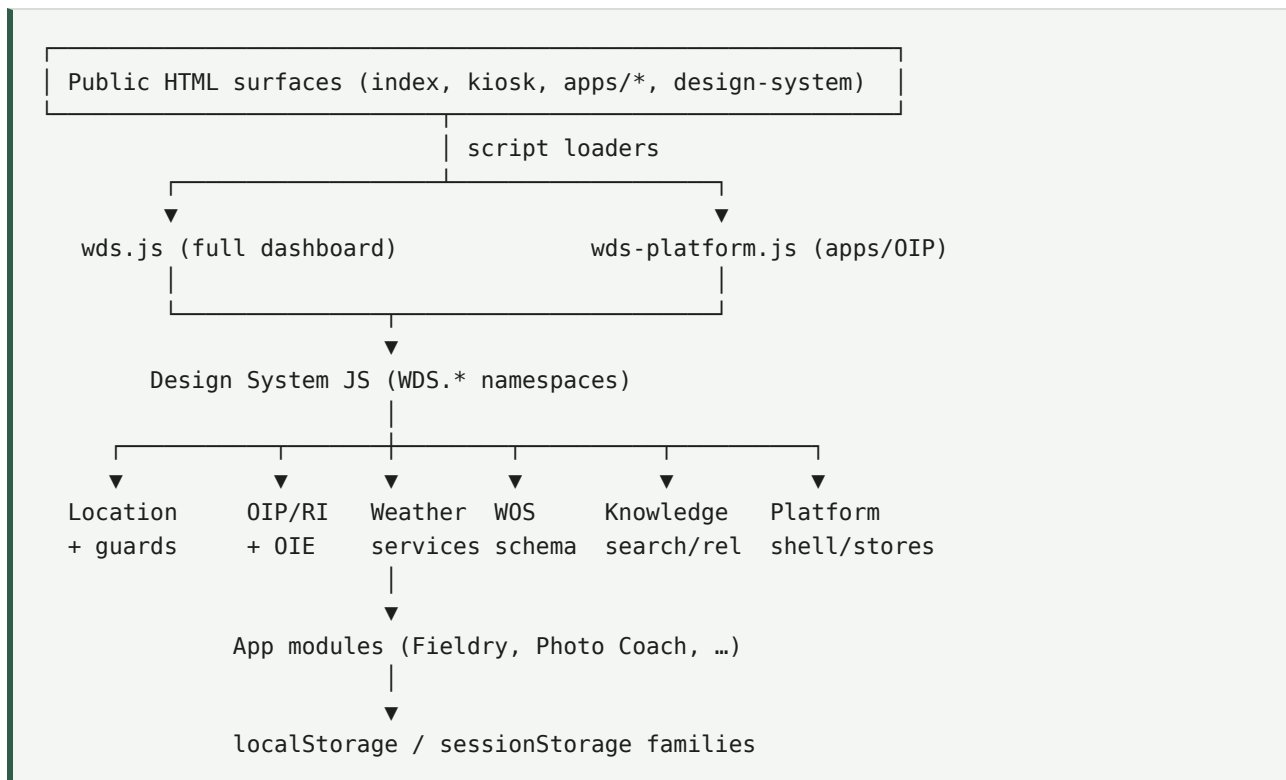
Healthy shared kernel, improved reliability posture, residual architectural debt that is documented rather than hidden.

Platform health is best described as **B+ / shippable**: critical silent-geography and boot bugs are closed; test gates are stronger; privacy defaults remain private-by-default; sample Knowledge data is honestly labeled.

Remaining risk concentrates in dual storage families, dual regional-intelligence stacks, ops HTML/JSON git churn, and national editorial honesty outside Pike-indexed content.

Repository Overview

High-level architecture



Content regionalization uses the Content Engine under `design-system/content-engine/` with county/region indices. Live-engine publish artifacts live under `data/` and ops pages (`status.html` , `debug.html`)—these must not drive user-facing weather/river coordinates.

Shared Platform Foundation

Commits: `3a28bc1` and subsequent adoption.

Module	Path	Role
Catalog	<code>design-system/js/platform/wds-platform-catalog.js</code>	Product registry for nav
Shell	<code>design-system/js/platform/wds-platform-shell.js</code>	Shared topbar/footer mount markers
Stores	<code>design-system/js/platform/wds-platform-stores.js</code>	Profile, locations, collections, settings

Module	Path	Role
Foundation UI	<code>design-system/js/platform/wds-platform-foundation.js</code>	Landing renderer for unfinished products
Future data	<code>design-system/js/platform/wds-platform-future-data.js</code>	Disabled APIs / GIS / licensing hooks

Docs: `docs/UNIFIED-PLATFORM.md` , `docs/PLATFORM-ARCHITECTURE.md` .

Knowledge Platform

Commit: `3900125` .

- Schema + domains + index + relationships under `design-system/knowledge/`
- Runtime: `wds-knowledge-core.js` , `wds-knowledge-search.js` , `wds-knowledge-relationships.js`
- Public API: `WDS.knowledge` , `WDS.knowledge.search` , `WDS.knowledge.related`
- Demo bundle is representative sample data, not production completeness
- Species essays remain in WSKB; Knowledge entries may link via `wskbId`

Docs: `docs/WAYPOINT-KNOWLEDGE-PLATFORM.md` .

Waypoint Observation Schema (WOS)

- Schema: `design-system/observations/schema-v1.json`
- Runtime: `wds-wos-core.js` + `wds-wos-extensions.js`
- App-specific data belongs in `observation.extensions[appId]`
- Privacy / location precision enums live on the observation
- Taxon id sources include `wskb` and `knowledge`

Docs: `docs/WAYPOINT-OBSERVATION-STANDARD.md` .

Application architecture

App	Path	Role
Dashboard	<code>index.html</code> + <code>js/home-boot.js</code>	Outdoor Intelligence home
Kiosk	<code>kiosk.html</code> + <code>js/kiosk*.js</code>	Full-screen live display
ForageCast	<code>apps/foragecast/</code>	Seasonal / habitat lab
Fieldry	<code>apps/fieldry/</code>	Life list + WOS notebook

App	Path	Role
Scenes landing	<code>apps/scenes/</code>	Photography product entry
Scene Builder	<code>apps/waypoint-scenes/</code>	Scene studio
Photo Coach	<code>apps/photo-coach/</code>	Coaching + profile
Sheds	<code>apps/shed-hunting/</code>	Foundation
Steepleaf	<code>apps/steepleaf/</code>	Foundation
SignalTerrain	<code>apps/signalterrain/</code>	Foundation
Savant	<code>apps/savant-sommelier/</code>	Foundation
Terrainbound	<code>apps/terrainbound/</code>	Retired redirect → Fieldry

Shared services

- **Location:** `wds-location.js` (coordinates-first, no silent fake “your location” for geo denial after prior hardening work; defaults only when user explicitly chooses)
- **Platform guard:** `wds-platform-guard.js` strips engine-classified packages from user surfaces
- **Weather / alerts / AQI / elevation / USGS water:** modular services under `design-system/js/weather/` and `water/`
- **Regional Intelligence:** engine + v2 core (both still loaded—debt)
- **Outdoor Intelligence Platform:** `wds-oip-*` package assembly
- **App boot:** `wds-app-boot.js` shared location + OIP bootstrap helper
- **Ethics / research integrity / provenance:** shared footnotes for scientific honesty

Audit Scope

This sprint reviewed the following areas end-to-end (read/inspect, then fix highest-value defects):

Shared platform

- Catalog, shell mount markers, stores, foundation renderer, future-data gate
- Loaders `wds.js` and `wds-platform.js` dependency order
- Repeated-load safety of `WDS.*` namespaces
- Navigation consistency across core instruments

Every application

- Dashboard, kiosk, status, debug
- ForageCast, Fieldry, Scenes, Scene Builder, Photo Coach (+ profile + guide)
- Sheds, Steepleaf, SignalTerrain, Savant Sommelier
- Terrainbound redirect
- Species profile and major design-system demos (asset path resolution)

Shared design system

- CSS entry points (`wds.css` , dashboard CSS, base components)
- Shared buttons, focus, skip link
- Gallery / photography helpers used by Scenes

Routing

- Hash routes (Fieldry SPA)
- Static HTML routes and meta-refresh redirect
- Smoke coverage of public paths

Storage

- Inventory of localStorage/sessionStorage families
- Migration markers and legacy keys
- Malformed payload behavior (Fieldry)

Knowledge Platform

- Schema, domains, demo bundle, search, relationships, WSKB delegation
- Unavailable / offline honesty in Fieldry capture

WOS

- Normalize/extensions preservation
- Privacy levels and location precision
- Fieldry extension namespace

Documentation

- UNIFIED-PLATFORM, ROADMAP, Knowledge, WOS, Fieldry MVP, architecture

- Gaps filled with STORAGE-INVENTORY and this audit

Tests

- All `automation/test-*.mjs` unit suites runnable without browser
- Validators under `scripts/validate-*.mjs`
- Headless Chrome smoke (`automation/smoke-browser.mjs`)
- GitHub Actions `smoke.yml`

Build / deployment

- `wds-build.js` version query / script tracking
- Live-engine artifact generation (not re-architected this sprint)
- `.gitignore` gaps for local churn

Accessibility

- Shared focus-visible, skip link, touch targets
- Form labels and status regions in Fieldry (prior MVP) and location prompt

Performance

- Duplicate listeners, dual RI stacks, Knowledge demo preload behavior
- No bundler migration (explicitly out of scope)

Privacy

- Private-by-default observations and platform profile
- Coordinate display gating in Fieldry
- Engine publish coordinates rejection

Security

- Secret scanning (no committed cloud keys found)
- XSS hotspots in `innerHTML` construction
- Future-data / marketplace disabled

Out of scope / not claimed: Formal penetration test, WCAG certification audit, every viewport on physical devices, backend auth, cloud sync, marketplace enablement.

Findings

Critical

C1 — Fieldry and ForageCast permanently null `WDS.appBoot`

Field	Detail
Description	App boot modules captured <code>global.WDS.appBoot</code> once at script evaluation time. Under <code>defer + dynamic</code> <code>wds-platform.js</code> chain, <code>appBoot</code> was not yet defined, so <code>boot</code> stayed <code>null</code> forever.
Impact	Location/OIP bootstrap always rejected; Fieldry/ForageCast ran without shared location intelligence.
Root cause	Eager binding incompatible with deferred platform loader ordering. Photo Coach already used lazy <code>getBoot() + waitForAppBoot()</code> .
Resolution	Rewrote both boots to lazy-create and wait for <code>WDS.appBoot</code> (Photo Coach pattern).
Files affected	<code>apps/fieldry/js/fieldry-boot.js</code> , <code>apps/foragecast/js/foragecast-boot.js</code>

C2 — WOS core hard-replace wiped extensions API

Field	Detail
Description	<code>wds-wos-extensions.js</code> could attach <code>WDS.observations.extensions</code> before <code>wds-wos-core.js</code> assigned a brand-new <code>WDS.observations = {...}</code> object, dropping <code>.extensions</code> . Fieldry also loaded extensions as a sibling while core arrived via platform loader.
Impact	Intermittent loss of <code>setExtension / getExtension</code> ; Fieldry extension writes could fail depending on race.
Root cause	Hard assignment instead of merge; platform loader omitted extensions script.
Resolution	Core now <code>Object.assign</code> s onto existing observations object and preserves <code>.extensions</code> . <code>wds-platform.js</code> loads <code>observations/wds-wos-extensions.js</code> immediately after core.
Files affected	<code>design-system/js/observations/wds-wos-core.js</code> , <code>design-system/js/wds-platform.js</code>

C3 — Kansas bounding box treated real users as “test coordinates”

Field	Detail
Description	<code>isKnownTestCoords</code> rejected any cache with <code>lat ∈ [37,40]</code> and <code>lng ∈ [-102, -94]</code> —most of Kansas—in addition to the engine publish point.
Impact	Legitimate Kansas users could have valid caches cleared / treated as stale; conflated engine artifact rejection with geography.

Field	Detail
Root cause	Over-broad heuristic intended to catch engine US-center leakage.
Resolution	<code>isKnownTestCoords</code> now only delegates to <code>isEnginePublishPoint</code> ($\approx 39.8283, -98.5795$).
Files affected	<code>design-system/js/wds-location.js</code>

High

H1 — OIP `get()` catch silently substituted Pike County

Field	Detail
Description	On failure, OIP built a package from <code>buildFallbackLocationState()</code> (default region Pike County, PA).
Impact	Users could see Pike editorial/coords after errors without understanding it was fallback geography.
Root cause	Fail-soft design that preferred “something to render” over honesty.
Resolution	Catch now returns an unavailable empty package (<code>meta.unavailable</code> , null coords, <code>blockStatus unavailable</code>)—no Pike substitution.
Files affected	<code>design-system/js/outdoor-intelligence/wds-oip-service.js</code>

H2 — Location resolve silently defaulted to Pike / engine default

Field	Detail
Description	<code>resolveLocation</code> (RI engine + OIP location) fell back to <code>defaultState</code> / <code>buildFallbackLocationState</code> whenever no coordinates existed.
Impact	Silent default geography for callers that omitted location.
Root cause	Historical “always return a region” contract.
Resolution	Defaults only when <code>request.allowDefaultLocation === true</code> . Otherwise return <code>unavailable</code> location state.
Files affected	<code>design-system/js/regional-intelligence/wds-regional-intelligence-engine.js</code> , <code>design-system/js/outdoor-intelligence/wds-oip-location.js</code>

H3 — Knowledge core hard-replace dropped `search` / `related`

Field	Detail
Description	Search/relationships modules attach methods onto <code>WDS.knowledge</code> . Reloading core replaced the object and could drop satellites.

Field	Detail
Impact	<code>WDS.knowledge.search / .related</code> could vanish after reload order quirks.
Root cause	Hard assignment without re-attach.
Resolution	Core merge-assigns and re-wires from <code>knowledgeSearch / knowledgeRelationships</code> if present.
Files affected	<code>design-system/js/knowledge/wds-knowledge-core.js</code>

H4 — Duplicate `location.onChange` listeners on reload

Field	Detail
Description	OIP service and RI engine registered <code>WDS.location.onChange</code> refresh handlers without once-guards.
Impact	Repeated script evaluation → duplicate fetches / refresh storms.
Root cause	No idempotent subscribe flag.
Resolution	Once-guards (<code>OIP._locationBound</code> , <code>global.__wdsRiEngineLocBound</code>).
Files affected	<code>wds-oip-service.js</code> , <code>wds-regional-intelligence-engine.js</code>

Medium

M1 — Reflected XSS risk in species profile boot

Field	Detail
Description	<code>?id=</code> query value concatenated into <code>innerHTML</code> on not-found path.
Impact	Potential HTML injection via crafted URL.
Root cause	Missing escape helper.
Resolution	Added <code>escapeHtml(id)</code> before interpolation.
Files affected	<code>design-system/species/profile-boot.js</code>

M2 — Unescaped `photo.id` in Scenes photography attributes

Field	Detail
Description	Titles/captions escaped; <code>data-photo-id</code> attributes used raw <code>photo.id</code> .
Impact	Attribute injection if ids ever become untrusted.
Root cause	Incomplete escaping discipline.

Field	Detail
Resolution	Wrap ids with existing <code>escapeHtml</code> .
Files affected	<code>apps/waypoint-scenes/js/photography.js</code>

M3 — Platform unit tests outside CI; smoke route list incomplete

Field	Detail
Description	Foundation/Knowledge/Fieldry/etc. tests existed locally but were not in <code>smoke.yml</code> ; smoke omitted Fieldry, ForageCast, foundations, Terrainbound.
Impact	Regressions could merge without CI signal.
Root cause	CI grew slower than the test inventory.
Resolution	Expanded <code>smoke.yml</code> and <code>smoke-browser.mjs</code> page list.
Files affected	<code>.github/workflows/smoke.yml</code> , <code>automation/smoke-browser.mjs</code> , <code>automation/test-platform-hardening.mjs</code>

M4 — Generated engine / status file churn in working tree

Field	Detail
Description	<code>data/live.json</code> , <code>health.json</code> , <code>status.html</code> , <code>debug.html</code> , etc. frequently dirty and historically risk committing Kansas engine snapshots.
Impact	Noisy diffs; accidental commits of ops artifacts.
Root cause	Tracked publish artifacts without strong local ignore strategy.
Resolution	Expanded <code>.gitignore</code> for local override patterns; documented in storage/audit docs. Did not untrack already-tracked files this sprint (ops publish model still uses them).
Files affected	<code>.gitignore</code> , docs

Low

L1 — Dashboard favorites dual-read

Field	Detail
Description	Favorites may be read from <code>waypoint-dashboard-widgets-v4</code> and legacy <code>waypoint-dashboard-favorites-v1</code> .
Impact	Dual source of truth; mild consistency risk.
Root cause	Incremental migration incomplete.

Field	Detail
Resolution	Documented; not consolidated this sprint (data-loss risk).
Files affected	<code>wds-dashboard-widget-data.js</code> (read path; unchanged)

L2 — Photo Coach legacy + growth storage families

Field	Detail
Description	Parallel keys for legacy coach vs entity repository / profile / coaching memory.
Impact	Confusion; migration complexity.
Root cause	Growth model layered without full cutover.
Resolution	Explicitly retained; inventory documented.
Files affected	Photo Coach repositories (unchanged this sprint)

L3 — Dual Regional Intelligence v1 + v2 in loader

Field	Detail
Description	Both stacks load via <code>wds.js</code> .
Impact	Bundle weight; dual fallbacks historically.
Root cause	Incremental RI rewrite.
Resolution	Documented debt; no loader surgery this sprint.
Files affected	<code>design-system/js/wds.js</code> (unchanged)

L4 — Terrainbound residue in education maps / tokens

Field	Detail
Description	Retired product still appears in some education/registry accents.
Impact	Mild discoverability confusion (redirect itself is correct).
Root cause	Incomplete residue cleanup.
Resolution	Deferred; redirect + catalog retired tier are correct.
Files affected	Various education maps (unchanged)

Stability Improvements

Initialization

- Fieldry/ForageCast wait for `WDS.appBoot` before creating boot instances.
- Hardening tests assert source patterns (`getBoot` , `waitForAppBoot`) and absence of eager null capture.

Routing

- Terrainbound meta-refresh + fallback links to Fieldry verified in smoke (lands on Fieldry title).
- Smoke route inventory expanded to cover foundations and primary instruments.

Module loading

- `wds-platform.js` now includes WOS extensions after core.
- WOS and Knowledge use merge-assign + preserve/re-attach satellites.
- Once-guards prevent stacked location listeners.

Storage

- Fieldry migration v2 remains idempotent; corrupt JSON does not crash `list()` .
- Storage inventory documented for all major families.
- No destructive consolidations performed.

Migration

- Runtime location migration continues to clear known stale engine/weather keys.
- Fieldry maps legacy `observationType` → life-list category when missing.
- Photo Coach migrations left dual-read (safe).

Caching

- Engine publish coordinates still rejected for user location caches.
- Kansas bbox no longer invalidates real caches.
- No service worker present (confirmed); cache-busting via `?v=` on dashboard loader path.

Error handling

- OIP failures → unavailable package with telemetry and blockStatus, not fake region.
- Location resolve → unavailable unless explicit default opt-in.

- Fieldry Knowledge search already surfaces offline/unavailable messaging (MVP).

Retry logic

- Not redesigned this sprint. Existing provider retries remain bounded by prior services.
- Duplicate refresh amplification from stacked listeners was reduced (once-guards).

Graceful degradation

- Per-module `blockStatus` on unavailable OIP packages.
- Dashboard platform guard continues to strip engine packages (“Refreshing local conditions...” when applicable).
- Knowledge optional for Fieldry capture (manual/unidentified still works).

Race-condition fixes

- AppBoot race (C1)
- WOS extensions wipe race (C2)
- Knowledge satellite drop (H3)
- Listener stacking (H4)

Duplicate code removed / avoided

- Did not rewrite kiosk onto `wds-app-boot` this sprint (larger behavioral risk); documented as debt.
- Did not delete RI v1.
- Prefer shared Photo Coach boot pattern over new abstractions.

Dashboard Review

Outdoor Intelligence

- Primary surface: `index.html` + dashboard engine modules.
- User package path is coordinates-first OIP (`liveFeedSource: user-oip` in smoke).
- Smoke (2026-07-12): homepage hydrated; weather/alerts/AQI/elevation/usgsWater/trailConditions reported live; **no Kansas river leak** on homepage.

Kiosk

- `kiosk.html` boot completes (`bootDone` , `notLocating` in smoke).
- Still related to live-engine display concerns; platform guard / normalize path validated by scripts.
- Dual boot lineage (`wds-app-boot` vs `kiosk-boot`) remains architectural debt.

Location detection

- `wds-location.js` v3 storage; geo + IP + manual county/state search.
- Explicit default region button remains user-initiated (Pike / indexed default)—acceptable.
- Silent defaults on resolve/get failures removed.

Weather

- Modular weather service; validators confirm photography/weather inputs use user coords.
- Remaining opportunity: in-flight request dedupe under load.

Rivers

- USGS gauges constrained; location-sensitive tests reject Kansas gauges for Pike users and accept local gauges.
- Homepage smoke showed no Kansas river leak.

Sunrise / sunset

- Daylight UTC harness PASS (Montgomery NY windows; regression times absent).
- Homepage smoke showed sunrise/sunset text for resolved location.

AQI

- Part of OIP block status; smoke showed live AQI on homepage during audit run (environment-dependent).

Photography conditions

- Canonical photography source is user-OIP (surface consistency validators PASS).
- Photo Coach / kiosk aligned to canonical module (no `live.json` photography fetch in coach).

Shared platform integration

- Dashboard uses full `wds.js` (includes platform shell/stores/future-data).

- Catalog-driven nav marks Fieldry/Scenes/etc.

Remaining concerns

1. National educational content still Pike-centric for distant coords (honesty/trust UX).
 2. Ops `status.html` may legitimately show Kansas engine content—do not confuse with user dashboard.
 3. Kiosk consolidation onto shared boot still pending.
 4. Widget favorites dual-read pending cleanup.
-

Knowledge Platform Review

Architecture

Shared reference backbone: schema → domains → records → relationships → search API. Apps query Knowledge instead of copying encyclopedias. Species long-form stays in WSKB.

Schema

- `design-system/knowledge/schema-v1.json`
- Domains include fieldry, forage, sheds, steepleaf, etc. (`domains.json`)
- Entries carry names (common/scientific/aliases), categories, tags, taxonomy, geography, citations, optional `wskbId`

Search

- `WDS.knowledge.search(query, options)` with domain/category/geo filters
- Deterministic scoring over names, aliases, tags, categories, keywords
- Fieldry capture uses `{ domain: "fieldry", category }`

Relationships

- Graph in `relationships.json`
- `related(id)` BFS neighbors; `path(from, to)` short ecology chains
- Loop protection via visited set in BFS

WSKB integration

- `resolveSpeciesDetail` loads WSKB when `wskbId` present

- WOS `taxonIdSource` allows `wskb` and `knowledge`

Sample data

- `samples/demo-bundle.json` (~21 representative entries)
- UI must label as sample (Fieldry does)
- Not production species completeness

Future expansion

- Domain packs per product
- Richer citations / seasonality
- Sensitive-species flags for auto-obscuring (designed, not fully automated)

Remaining work

- Expand beyond demo honesty without implying completeness
- Ensure all consuming apps tolerate Knowledge unavailable (Fieldry does; foundations less dependent)
- Relationship coverage growth

Observation Platform Review

Schema

WOS v1 research-grade observation package: meta, observer, taxon, observedAt, location (+ privacy precision), habitat, context, record, media, verification, research, license, privacy, revisions.

Extensions

```
observation.extensions[appId] = { schemaVersion, updatedAt, ...payload }
```

Fieldry uses `extensions.fieldry` (category, unidentified, tags, knowledgeId, privacyLevel, ...) mirrored into `meta.fieldry` for resilience.

`normalizeObservation` shallow-copies extension objects per appId (fixed earlier in Fieldry sprint; merge-load fixed in hardening).

Privacy

- Extension privacy levels: `private` | `shared` | `public` | `anonymized` (default private in Fieldry)
- Location precision: `exact` | `obfuscated` | `county` | `hidden`
- `publicCoordinates()` respects precision
- Fieldry UI never shows exact coords when precision is regional/hidden

Storage

- Fieldry: `waypoint-fieldry-observations-v1` on device
- No cloud sync in MVP
- Export JSON/CSV for personal archive

Migrations

- Marker `waypoint-fieldry-migration-v2`
- Idempotent enrichment; malformed payloads preserved
- Legacy `observationType` → category mapping

Future scalability

- Shared WOS across apps is the correct path
- Avoid per-app observation schemas
- Sensitive-species auto-obscuring and media blobs are next-layer work
- Account sync reserved in retention enums but not enabled

Application Review

Dashboard

Aspect	Detail
Maturity	Live production surface
Completed	OIP integration, widgets, briefing, location pipeline, platform guard, validators
Remaining MVP	National-mode honesty UX outside Pike zone; favorites single source of truth
Technical debt	Dual RI stacks; widget favorites dual-read; large dashboard bundle
Recommendations	Next focus: national honesty + weather coalescing; keep engine artifacts off user paths

Waypoint Scenes (Scene Builder + landing)

Aspect	Detail
Maturity	Live
Completed	Scene studio, photography gallery, outdoor context bridge, XSS id escaping
Remaining MVP	Continued polish; ensure all dynamic HTML remains escaped
Technical debt	Some effect defs rendered via innerHTML (static today)
Recommendations	Keep attribute escaping discipline; avoid data-driven unsanitized HTML

Photo Coach

Aspect	Detail
Maturity	Live public product (critique, batch, profile, personalized coaching)
Completed	Growth models, profile engine, coaching memory, correct lazy boot
Remaining MVP	Storage consolidation UX clarity; collections polish
Technical debt	Legacy vs growth localStorage families
Recommendations	Dedicated migration sprint with dual-read + user-visible confirmation; do not delete legacy keys opportunistically

Fieldry

Aspect	Detail
Maturity	Usable Life List MVP
Completed	Home, capture, Knowledge suggestions, unidentified records, history, life list, stats, achievements, privacy, collections hooks, boot fix
Remaining MVP	Media attach UI; sensitive-species auto-obscure; richer Knowledge packs
Technical debt	Dual write meta.fieldry + extensions.fieldry; migration-on-read
Recommendations	Media + obscuring sprint after platform stability; keep non-competitive tone

ForageCast

Aspect	Detail
Maturity	Live seasonal/habitat lab
Completed	Shared boot (now fixed), season table, OIP context

Aspect	Detail
Remaining MVP	Daily recommendation package (roadmap)
Technical debt	Ensure all entry points use lazy boot if additional scripts added
Recommendations	Product deepening on recommendations without forking location/OIP

Sheds

Aspect	Detail
Maturity	Foundation landing + models
Completed	Find/species models, foundation JSON, private-by-default finds store
Remaining MVP	Personal finds UI with private coordinates
Technical debt	Scripts may lack ?v= cache bust
Recommendations	Build finds UI on shared shell + WOS extensions when capture deepens

Steepleaf

Aspect	Detail
Maturity	Foundation (tea/brew models)
Completed	Landing, stores, Knowledge domain samples
Remaining MVP	Brew session UX
Technical debt	Same foundation cache-bust gap
Recommendations	Grow from Knowledge tea domain; avoid parallel encyclopedia

SignalTerrain

Aspect	Detail
Maturity	Foundation (receiver/incident models)
Completed	Landing + local stores
Remaining MVP	Receiver management UI
Technical debt	Foundation-level only
Recommendations	Keep radio domain in Knowledge; private incident locations

Savant Sommelier

Aspect	Detail
Maturity	Foundation (site/winery/wine)
Completed	Landing + models + Knowledge wine/soil samples
Remaining MVP	Tasting/site workflows
Technical debt	Foundation-level only
Recommendations	Shared shell; no marketplace features

Storage Audit

Canonical companion doc: `docs/STORAGE-INVENTORY.md` . Consolidated table:

Storage key	Owner	Schema / version	Migration behavior	Status
<code>waypoint-platform-profile-v1</code>	platform stores	Profile v1	create-on-load	Current
<code>waypoint-platform-locations-v1</code>	platform stores	Locations[]	overwrite save	Current
<code>waypoint-platform-collections-v1</code>	platform stores	Collections[]	favorites helper	Current
<code>waypoint-platform-settings-v1</code>	platform stores	Settings	patch merge	Current
<code>wds-location-v3</code>	wds-location	Location state	migrate from v1/v2 clear	Current
<code>wds-location-v2 / v1</code>	legacy	—	cleared by runtime migration	Legacy
<code>wds-location-prompted</code>	wds-location	flag	—	Current
<code>waypoint-runtime-migration</code>	runtime migration	marker	idempotent purge	Current
<code>waypoint-active-build</code>	wds-build	commit id	—	Current
<code>waypoint-briefing-snapshot-v1</code>	briefing	cache	—	Current
<code>waypoint-outdoor-context-v1</code>	ecosystem bridge	sessionStorage	—	Current
<code>waypoint-dashboard-widgets-v4</code>	dashboard	layout+favorites	from v1–v3	Current
<code>waypoint-dashboard-widgets-v1 ... v3</code>	dashboard	—	migrated	Legacy
<code>waypoint-dashboard-favorites-v1</code>	dashboard	favorites	dual-read	Legacy

Storage key	Owner	Schema / version	Migration behavior	Status
waypoint-fieldry-observations-v1	Fieldry	WOS[]	migration v2 enrich	Current
waypoint-fieldry-device-id	Fieldry	device id	create once	Current
waypoint-fieldry-migration-v2	Fieldry	marker	idempotent	Current
waypoint-photo-records-v1	Photo Coach growth	PhotoRecord[]	—	Current
waypoint-photo-shoots-entity-v1	Photo Coach growth	Shoot[]	—	Current
waypoint-photographer-profile-v1	profile engine	profile	migrate helpers	Current
waypoint-photo-coaching-memory-v1	coaching	memory	—	Current
waypoint-photo-coaching-prefs-v1	coaching	prefs	—	Current
waypoint-photo-coach-journey-v1	Photo Coach UI	progress	—	Current
waypoint-photo-coach-profile-v1	legacy coach	profile	retained	Legacy
waypoint-photo-coach-sessions-v1	legacy coach	sessions	retained	Legacy
waypoint-photo-coach-shoots-v1	legacy coach	shoots	retained	Legacy
waypoint-sheds-finds-v1	Sheds	finds	—	Current
waypoint-steepleaf-teas-v1	Steepleaf	teas	—	Current
waypoint-steepleaf-brews-v1	Steepleaf	brews	—	Current
waypoint-signalterrain-receivers-v1	SignalTerrain	receivers	—	Current
waypoint-signalterrain-incidents-v1	SignalTerrain	incidents	—	Current
waypoint-savant-wineries-v1	Savant	wineries	—	Current
waypoint-savant-wines-v1	Savant	wines	—	Current
waypoint-savant-sites-v1	Savant	sites	—	Current
waypoint-wskb-recent-v1	WSKB	recent	—	Current
Debug keys (waypoint-debug-location , waypointDebugSnapshot)	debug tools	—	—	Ops

Rules: never silently erase valid history; prefer dual-read; migrations idempotent; malformed JSON quarantine/preserve.

Security & Privacy Review

Secrets review

- No committed `sk-` , `AIza` , `ghp_` , `AKIA` , or PEM private keys found in audit pass.
- eBird and commercial weather keys are environment/runtime-config only.
- `.env` under automation week-away is gitignored.

Privacy protections

- Platform profile default private.
- Fieldry observations default `privacyLevel: private` , location precision default `county` (regional).
- No public social feeds, followers, or leaderboards in Fieldry MVP.
- Future-data marketplace/licensing APIs return disabled.

Coordinate handling

- Engine publish point rejected for user caches.
- Fieldry `formatLocation` suppresses exact coords unless precision is `exact` .
- OIP no longer injects Pike coordinates on failure.
- Darwin Core / public coordinate helpers respect precision enums.

Future data safeguards

```
WDS.futureData.ENABLED === false
WDS.futureData.enable() → { ok: false, enabled: false, ... }
```

Hard gate until product + legal review.

Remaining concerns

- Client-only enforcement (anyone with DevTools can read `localStorage`)—expected for this architecture; cloud sync must not weaken defaults.
- Ops pages expose engine diagnostics by design—keep them non-public or clearly ops-only in deployment.
- `innerHTML` remains a recurring footgun; escaping is mostly disciplined but requires review on every new UI.

Accessibility Review

Improvements this sprint

- Shared CSS: `.wds-btn` and `.ws-topnav` a minimum height 44px; `.wds-btn--sm` 36px (`wds-base.css`).
- Existing `:focus-visible` outline and `.wds-skip` retained.
- Location prompt already uses dialog semantics + `aria-live` status.
- Fieldry MVP (prior commit) includes labels, status regions, reduced-motion CSS.

Remaining issues

- Not a full WCAG audit of every page.
- Some foundation landings may lack exhaustive keyboard dialogs (simple pages).
- Kiosk/status ops UIs are display-oriented; a11y priority is user instruments.
- Color contrast not instrumented automatically this sprint.

Recommendations

1. Add axe/lighthouse CI sample on homepage + Fieldry + Photo Coach.
2. Audit modal focus traps in Scenes gallery/Photo Coach.
3. Ensure all new forms use visible labels (no placeholder-only).

Performance Review

Improvements

- Once-guards stop duplicate OIP/RI refresh subscriptions (reduces fetch amplification).
- No large new bundles introduced by hardening.

Remaining bottlenecks

- `wds.js` loads a very large ordered script list (dashboard cold start).
- Dual RI v1+v2.
- Fieldry preloads Knowledge demo bundle (acceptable for MVP; watch size as packs grow).
- Foundation apps often load platform scripts without `?v=` (stale cache risk more than CPU).

Future opportunities

- Split dashboard widgets into lazy chunks (without framework migration).
 - Deduplicate in-flight weather requests.
 - Remove RI v1 after call-site audit.
 - Knowledge: load domain packs on demand instead of full demo always.
-

Documentation Review

Updated

- `docs/UNIFIED-PLATFORM.md` — links to audit + storage inventory; last-updated notes in prior commits
- `docs/ROADMAP.md` — Fieldry status / next phase (Fieldry MVP commit)
- `docs/FIELDRY-LIFE-LIST-MVP.md` — product/technical MVP doc (Fieldry commit)

Added

- `docs/PLATFORM-AUDIT-2026-07.md` (this report; expanded)
- `docs/STORAGE-INVENTORY.md`

Still needed

- Runbook for live-engine publish vs user dashboard (ops vs product)
 - Photo Coach storage migration design doc before consolidation sprint
 - National-mode content honesty UX spec
 - Optional: ADRs for “no silent default location” and “extensions merge-load”
-

Tests

Unit / Node suites (2026-07-12 hardening verification)

Test	Result	Coverage / notes
<code>automation/test-platform-hardening.mjs</code>	PASS	WOS/Knowledge merge-load, KS coords, futureData, boot patterns, privacy display, corrupt storage, Terrainbound redirect, XSS escape

Test	Result	Coverage / notes
automation/test-platform-foundation.mjs	PASS	Catalog, shell, stores, extensions, futureData, foundations, Fieldry categories
automation/test-knowledge-platform.mjs	PASS	Schema, search, relationships, domains, WSKB link fields
automation/test-fieldry-mvp.mjs	PASS	Capture, life list, stats, achievements, privacy, migration
automation/test-photographer-profile.mjs	PASS	Tiers, niches, exclusions, corrections, privacy
automation/test-personalized-coaching.mjs	PASS	Growth narratives, outing plans, memory prefs
automation/test-profile-migration.mjs	PASS (when server available)	Persistent profile migration in browser context
automation/test-kiosk-modules.mjs	PASS (earlier full run)	Kiosk module integrity
automation/test-kiosk-location-boot.mjs	PASS (Chrome + server)	Includes stale Kansas cache rejection
automation/test-trail-browser.mjs / test-trail-conditions.mjs	Present	Trail surfaces; not all in CI

Validators

Script	Result
scripts/validate-location.mjs	PASS
scripts/validate-dashboard-data.mjs	PASS
scripts/validate-surface-consistency.mjs	PASS
scripts/validate-location-sensitive.mjs	PASS (daylight, rivers, weather coords, platform guard)

Headless smoke (automation/smoke-browser.mjs)

Result: SMOKE: PASS (local Chrome, 2026-07-12)

Routes exercised: homepage, kiosk, status, debug, ForageCast, Fieldry, waypoint-scenes, scenes, photo-coach, photo-coach-profile, sheds, steepleaf, signalterrain, savant, terrainbound-redirect, species-profile.

Observations:

- Homepage: no Kansas river leak; user-oip source; hydrated.
- Status page may flag Kansas content (ops engine page—expected).

- Fieldry/ForageCast may still show `aria-busy` briefly during location prompt at capture time; **no console errors**.
- Terrainbound redirect resolved to Fieldry title.

CI

`.github/workflows/smoke.yml` now runs smoke + profile migration + platform foundation + knowledge + fieldry + hardening + photographer profile + personalized coaching + validators.

Git Summary

Item	Value
Branch	<code>main</code>
Final hardening commit	<code>0b748364f4ece7c5ffef4ecc54fdf4e078ca9bbe</code>
Pushed to <code>origin/main</code>	Yes (<code>f26eb40</code> . <code>0b74836</code> at time of hardening push)
This report commit	See follow-up commit if created after expansion

Commits created during the broader July platform arc

1. `3a28bc1` — Unified Platform Foundation
2. `3900125` — Knowledge Platform
3. `f26eb40` — Fieldry Life List MVP
4. `0b74836` — Hardening (boot, location, XSS, CI, initial audit/storage docs)

Files changed in hardening commit `0b74836` (19 files)

`.github/workflows/smoke.yml` , `.gitignore` , Fieldry/ForageCast boots, `photography.js`, `smoke-browser`, `test-platform-hardening`, `wds-base.css`, `knowledge-core`, `wos-core`, `oip-location`, `oip-service`, `regional-intelligence-engine`, `wds-location`, `wds-platform`, `profile-boot`, `PLATFORM-AUDIT-2026-07.md`, `STORAGE-INVENTORY.md`, `UNIFIED-PLATFORM.md`

Working tree note (post-push)

Local dirty files typically remain for **generated/ops churn** and were **not** committed:

- `data/build-info.json` , `data/health.json` , `data/live.json` , `data/publish-state.json`
- `status.html` , `debug.html` , `kiosk.html`

- design-system demo/shell HTML, `wds-build.js`
- `waypoint-importer/assets/waypoint-importer.desktop`

Technical Debt

Prioritized:

1. **P1 — Photo Coach legacy vs growth storage consolidation** (data-loss risk if rushed)
2. **P1 — Dashboard favorites single source of truth**
3. **P1 — National-mode content honesty** outside Pike editorial zone
4. **P2 — Consolidate kiosk boot onto `wds-app-boot.js`**
5. **P2 — Remove or isolate RI v1** after call-site audit
6. **P2 — Weather in-flight request dedupe**
7. **P2 — Foundation script `?v=` cache busting**
8. **P2 — Untrack or segregate engine publish artifacts** from everyday git status
9. **P3 — Terrainbound residue cleanup** in education maps
10. **P3 — Fieldry dual meta/extensions mirror consolidation**
11. **P3 — Automated a11y CI**

Risks

Risk	Severity	Monitoring
Silent geography regressions	High (mitigated)	Keep location-sensitive validators + hardening tests in CI
Engine <code>live.json</code> leaking into user weather/rivers	High (guarded)	<code>validate-location-sensitive</code> , homepage smoke <code>hasKansasRiverLeak</code>
Accidental commit of Kansas engine snapshots	Medium	Review diffs; prefer ignore/untrack strategy
Photo Coach storage split user confusion	Medium	Document; migrate deliberately
Knowledge sample mistaken for completeness	Medium	Keep UI labels; expand docs
Large dashboard bundle regressions	Low–Medium	Performance budget optional

Risk	Severity	Monitoring
XSS via new innerHTML features	Medium	Code review checklist; escape helpers
Client-side privacy only	Accepted	Disclose; backend later

Recommended Next Sprint

Single highest-value next sprint: National-mode honesty + weather request coalescing + dashboard favorites cleanup

Why this over alternatives

- Location correctness bugs that caused wrong *coordinates* are largely closed; the remaining user-trust issue is **wrong editorial framing** (Pike-centric content presented as local for distant users).
- Weather coalescing reduces provider load and flaky UI under refresh storms—pairs naturally with dashboard work.
- Favorites dual-read is a small, high-clarity consistency win on the same surface.
- Photo Coach storage consolidation is valuable but higher data-loss risk—better as its own dedicated sprint with migration fixtures.
- Fieldry media attach is product expansion, not stability—lower priority immediately after a hardening sprint.

Success criteria (suggested)

1. Distant users see explicit “national educational” framing—not implied local Pike intelligence.
2. Weather service dedupes identical in-flight requests.
3. Favorites read/write only `widgets-v4` (legacy key read-only migrate-once).
4. Validators + smoke remain green.

Overall Assessment

Waypoint Studio after this audit is a **credible multi-product platform with a real shared kernel**, not a loose collection of demos. The July work closed the class of bugs that most damage trust: **wrong place, wrong boot, wiped APIs, and quiet fallbacks**. Test coverage and CI now better match the architecture’s importance.

The repository is **not** “done.” Foundations are intentionally incomplete; Photo Coach carries parallel storage; the dashboard bundle is heavy; national content honesty remains a product-engineering problem. Those are **known, documented debts**, not unknown footguns.

For a senior engineer joining tomorrow: trust `WDS.observations`, `WDS.knowledge`, `WDS.platform`, and coordinates-first OIP; never reintroduce silent Pike/Kansas defaults; keep future-data disabled; treat `data/live.json` as engine ops, not user truth; run the hardening + location validators before merging location-sensitive changes.

Engineering grade after sprint: shippable core instruments, disciplined foundations, improved reliability posture.

Appendix A — Verification commands

```
node automation/test-platform-hardening.mjs
node automation/test-platform-foundation.mjs
node automation/test-knowledge-platform.mjs
node automation/test-fieldry-mvp.mjs
node automation/test-photographer-profile.mjs
node automation/test-personalized-coaching.mjs
node scripts/validate-location.mjs
node scripts/validate-dashboard-data.mjs
node scripts/validate-surface-consistency.mjs
node scripts/validate-location-sensitive.mjs
# With static server + Chrome:
python3 -m http.server 8080 &
node automation/smoke-browser.mjs http://127.0.0.1:8080
node automation/test-kiosk-location-boot.mjs http://127.0.0.1:8080
```

Appendix B — Related documents

- `docs/UNIFIED-PLATFORM.md`
 - `docs/PLATFORM-ARCHITECTURE.md`
 - `docs/WAYPOINT-KNOWLEDGE-PLATFORM.md`
 - `docs/WAYPOINT-OBSERVATION-STANDARD.md`
 - `docs/FIELDRY-LIFE-LIST-MVP.md`
 - `docs/STORAGE-INVENTORY.md`
 - `docs/ROADMAP.md`
-

End of report.